

PROJECT PRESENTATION

RecallGPT

An AI-Powered Second Brain
Chat with your notes. Track tasks, expenses & memories.

[Problem](#) · [Solution](#) · [Architecture](#) · [Workflows](#) · [Demo](#) · [Roadmap](#)

Django + DRF

LangChain

RAG

pgvector

MiniLM Embeddings

Llama 3.1

Supabase

ChromaDB

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Agenda

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The Problem — Your Notes Remember Nothing

We capture our lives in notes — meetings, ideas, to-dos, spending, trips. But that knowledge is write-only: finding anything later means scrolling, and tracking tasks or expenses means retyping everything into other apps.

Search is keyword-blind

'What did I spend on transport?' finds nothing unless you literally wrote the word 'transport'. Meaning is lost; only exact words match.

Manual double-entry

Tasks go into a to-do app. Expenses into a spreadsheet. Photos into a gallery. The same information is entered three or four times.

Privacy vs. AI trade-off

Cloud AI assistants can read your notes — but personal journals, finances and photos are exactly the data you don't want to upload.

THE IDEA

What if your notes were a second brain you could simply talk to?

Upload plain markdown notes once. Ask questions in natural language and get answers grounded in your own writing. Tasks, expenses and photo memories are extracted automatically — no retyping, and nothing has to leave your machine.

What is RecallGPT?

A privacy-first personal knowledge assistant built with Django: upload your markdown notes, then converse with them using Retrieval-Augmented Generation (RAG) — while the system silently extracts tasks, expenses and image memories with zero manual data entry.

Conversational RAG

Ask natural-language questions over your notes.
Semantic retrieval + LLM answers grounded in your data.

Task Extraction

Auto-detects '- []', 'todo:', 'task:', 'action:' lines.
Pending vs. completed tracked in a dashboard.

Expense Tracking

Parses 'spent 500 on coffee', 'bought X for 350'...
Multi-currency: \$, ₹, €, £, Rs., INR, USD, EUR, GBP.

Image Memory

Gallery with EXIF capture (camera, time, location).
Semantic image recall woven into chat answers.

Multi-Session Chat

Independent 'notes' and 'general' chat sessions.
Full history persisted; resume any conversation.

Private by Design

Local LLM via Ollama — notes stay on the machine.
Per-user data isolation; NVIDIA cloud as fallback.

One input — a plain .md note — feeds every capability. Write once; recall, track and relive automatically.

← Back to home



RecallGPT

Welcome back

Sign in to continue working with your notes, chats, tasks, and image memories.

Username



Your username

Password



Your password



Sign in →

New to RecallGPT? [Create an account](#)

● AI memory index active

Your second brain, always in sync.

Everything you've captured — notes, tasks, expenses and photo memories — ready the moment you sign in.

24

Notes

08

Tasks

16

Images

- ✓ Chat with your own knowledge base
- ✓ Auto-extracted tasks & expenses
- ✓ Searchable visual memory gallery

Start free — no credit card

Build your own AI second brain.

RecallGPT keeps your notes, conversations, action items, expenses and visual memories in one focused, intelligent workspace.



Upload personal notes

Ask questions with note-aware context.



Track follow-through

Tasks and expenses extracted automatically.



Build image memory

Search photos by tags, dates and details.

← Back to home



RecallGPT

Create your account

Set up access to your personal AI workspace in seconds.

Username



Pick a username

Phone number



Your phone

Email



you@example.com

Password



Create a strong password



Use at least 8 characters with a mix of letters and numbers.

Create account →

Already registered? [Sign in](#)



RecallGPT

MAIN MENU



Dashboard



Chat with Notes



RecallGPT Chat



Task Overview



Expense Tracker



AI Image Studio



Memories Gallery



Upload



Sync DB

ACCOUNT



Light Mode



Logout



abhirup746

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Good evening , Abhirup746 · Thursday, July 16

RecallGPT

Your Intelligent AI Memory Workspace

Upload notes, track action items, organize expenses, and converse with your personal RAG assistant — everything you capture, understood and instantly recalled.



Start AI Chat



Upload Notes



5 indexed

5

Knowledge Base

Notes indexed & searchable



12 done · 0 left

12

Action Items

Tasks extracted from notes



↑ +3 this week

25

Conversations

Chat sessions started



₹3,566 left

₹6,434

Monthly Spend

Tracked from your notes



Productivity Overview

Live from your workspace



Today's Summary



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Dashboard



Chat with Notes



RecallGPT Chat



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Insights

Real data only

Task Completion

Done vs. remaining



100%
12 of 12
done

Weekly Activity

Chat sessions · last 7 days



Monthly Expenses

By day · this month



Recent Activity

Latest across your workspace



Jul 15, 14:13

AI chat: hey there



Jul 15, 11:30

AI chat: give me that picture of football world cup



Jul 11, 08:37

AI chat: do you rember the pic fifa world cup



Jul 07, 14:51

Notes chat: summarize it



Jul 07, 14:50

AI chat: hey there



Jul 07, 14:22



Quick Access



RecallGPT Chat

General-purpose AI assistant



Chat with Notes

RAG over your knowledge base



AI Image Studio

Generate visuals from a prompt



Tasks Board

Track & complete action items



Expenses Center

Ledger, budget & alerts



RecallGPT vs. Traditional Note & Photo Apps

Traditional apps (Keep, iCloud, files)	RecallGPT
Search Exact keyword matching — one missing letter means zero results.	Semantic search: understands concepts ('laptop' ≈ 'computer').
Task tracking Separate to-do apps; lists updated by hand.	Auto-extracted from markdown; ticking a task edits the file itself.
Expense tracking Manual entry into spreadsheets or finance apps.	Natural-language parsing builds expense reports automatically.
Image recall Search by filename or hand-typed tags only.	Auto-tags via NLP + EXIF camera/date/GPS metadata extraction.
AI integration None, or basic spell-check.	Llama 3.1 LLM answers grounded in vector-retrieved note context.
Storage Files on disk / server trees — paths break, hosts wipe files.	Custom DatabaseStorage keeps notes & media as database binaries.

RecallGPT unifies raw, unorganized text files and media into an active conversational memory.

RAG — Retrieval-Augmented Generation, in Plain Words

An LLM alone answers from generic training data — it has never seen your notes. RAG fixes this: first retrieve the few note passages relevant to the question, then let the LLM answer using only those passages as context.

1. Chunk

Notes are split into small overlapping passages

2. Embed

Each chunk becomes a 384-dim vector capturing its meaning

3. Index

Vectors stored per user in pgvector / ChromaDB

4. Retrieve

Question is embedded; top-K nearest chunks are fetched

5. Generate

LLM answers strictly from the retrieved context

Why 'embeddings' matter

An embedding is a list of numbers that encodes meaning: sentences about the same idea land close together in vector space, even with different words. So the query 'transport costs' finds the note line

"paid 1200 for petrol"

— no shared keyword required. RecallGPT uses the compact all-MiniLM-L6-v2 model, small enough to run on CPU.

Why grounding matters

The prompt template instructs the model to use only the supplied context. Every amount, task and date in the answer comes from the user's own note, not from the model's memory — which sharply limits hallucination and makes answers auditable.

This is the same architecture used by production document-QA systems.

Technology Stack

Web & API

Django 6.0	Web framework, ORM, auth, templates
Django REST Framework	JSON REST API, serializers, permissions
Gunicorn + WhiteNoise	Production WSGI server & static files

AI / LLM Layer

LangChain	RAG orchestration, QA chains, text splitting
Ollama (local)	llama-3.1-8b-instruct — private, on-device
NVIDIA AI Endpoints	ChatNVIDIA cloud fallback (same model family)

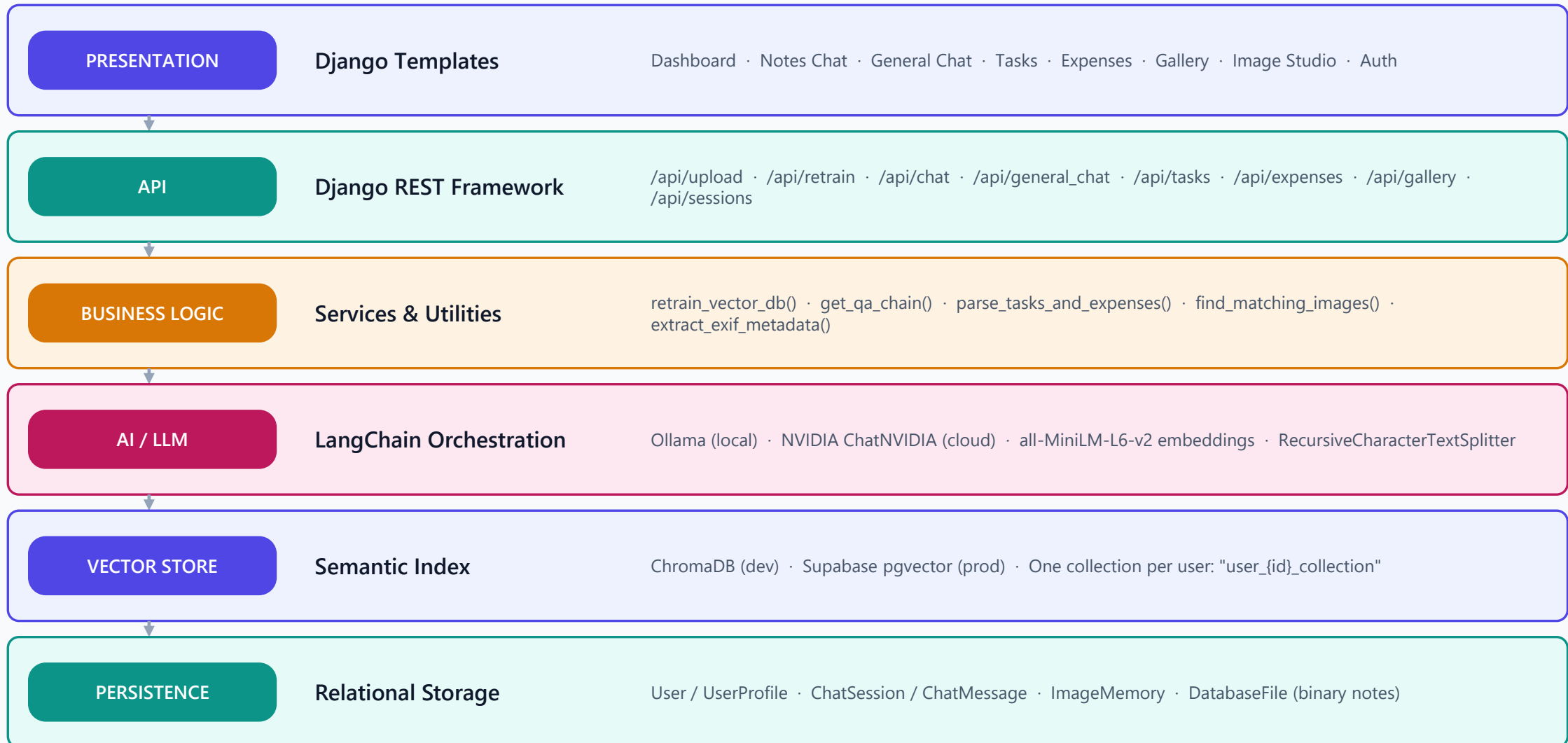
Embeddings & Vectors

Sentence Transformers	all-MiniLM-L6-v2 · 384-dim embeddings
ChromaDB	Local vector store for development
Supabase pgvector	Production vector search in PostgreSQL

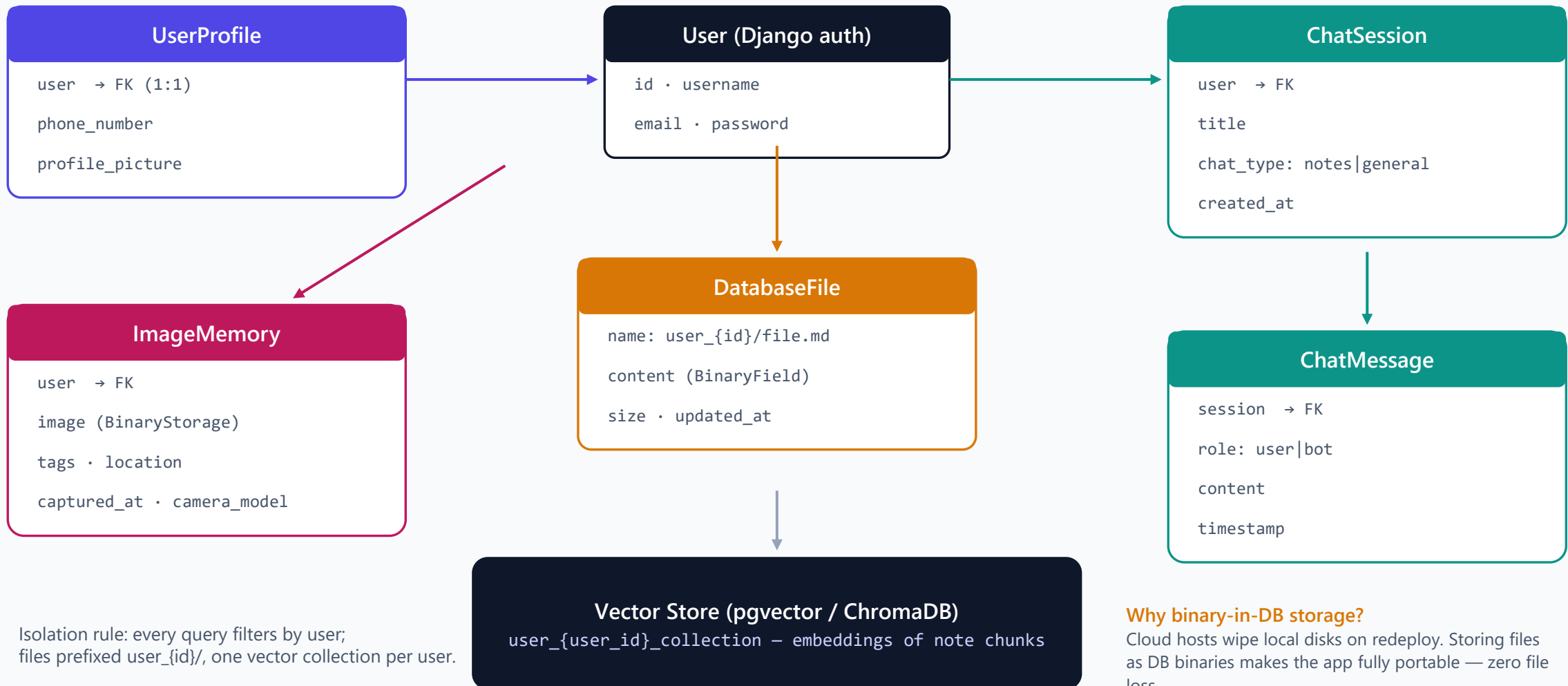
Data & Storage

PostgreSQL / SQLite	ORM models: sessions, messages, images
DatabaseFile (BinaryField)	Markdown notes stored as binary blobs
Pillow	Image processing & EXIF metadata extraction

System Architecture — Six Layers

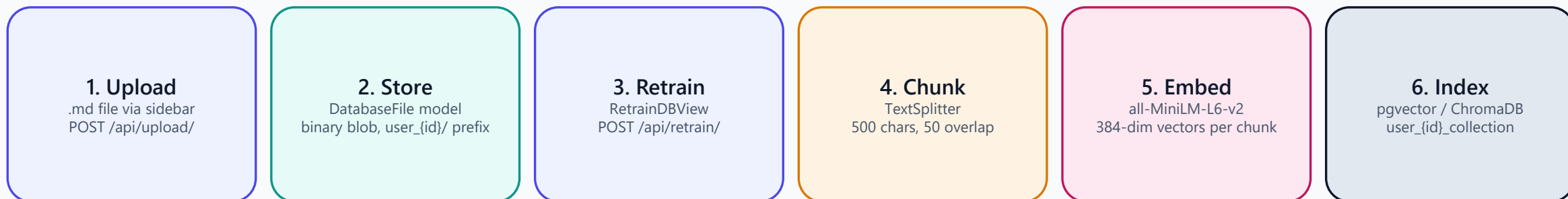


Data Model — Everything Scoped to the User



Note Upload & Vector Indexing

Turning raw markdown into a searchable semantic index — runs on every 'Retrain Vector DB' click.



Why chunk + embed?

LLMs can't read a whole notebook per question. Notes are split into small overlapping chunks; each chunk becomes a dense vector that captures its meaning. Similar meanings → nearby vectors → fast semantic lookup.

The embedding model is loaded once and cached in memory (~400 MB), protected by a threading lock against concurrent retrains.

Per-user collections

Every user gets an isolated vector collection —
user_7_collection, user_8_collection, ...

Retraining wipes and rebuilds only that user's collection, so uploads are reflected instantly without touching anyone else's index.
Dev uses local ChromaDB; production uses Supabase pgvector.

Notes Chat — The RAG Pipeline





RecallGPT

MAIN MENU



Dashboard



Chat with Notes



RecallGPT Chat



Task Overview



Expense Tracker



AI Image Studio



Memories Gallery



Upload



Sync DB

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+ New Chat

Recall my image of boy p...

2026-07-04T12:38:18.142582Z

Show me the cricket photo

2026-07-04T12:36:26.679160Z

What are my expenses?

2026-07-04T12:36:14.216716Z

Summarize my tasks

2026-07-04T12:36:07.123623Z

summarize it

2026-07-04T12:22:46.306933Z

summarize it

Ask questions based on your uploaded markdown notes.

summarize the notes

RECALLGPT

To summarize the notes:

- The user needs to review their monthly spend and categorize expenses, prepare an invoice summary, and send a payment reminder to a vendor.
- They have a to-do list that includes updating the cash flow forecast, checking the payroll submission, and finalizing budget projections for Q3.
- The user also needs to finish a quarterly report by Friday and send it to the client for review.
- In terms of expenses, they have paid \$120 for office supplies, ₹3,500 on travel, \$29 for a subscription, \$180 for utilities, \$1200 for rent, \$75 for parking, and \$12 for snacks. They have also received \$500 from a client payment.



Ask something about your notes...



Answers are based on your uploaded notes. Click to upload a new note or to select an existing one.

Task & Expense Extraction Engine

A deterministic regex engine (parsers.py) — no LLM cost, instant, and auditable. And it's two-way: ticking a task on the dashboard rewrites '[]' to '[x]' in the stored markdown; deleting an expense removes that line and retrains the index.



Task patterns		Expense patterns (multi-currency: \$ ₹ € £ Rs. INR USD EUR GBP)	
- [] Buy groceries	checkbox → pending task	spent 500 on coffee	narrative 'spent/paid'
- [x] Pay rent	checked → completed task	bought a book for 350	'bought ... for' format
todo: renew car insurance	prefix format	coffee cost 120	'cost' format
task: prepare slides	prefix format	- Dinner - 250 - Lunch: Rs. 150	bullet with separator
action: email the vendor	prefix format	- ₹250 - Food	amount-first bullet



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Chat with Notes



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Task Overview

Automatically extracted to-dos from your markdown notes.

+ Add Task

All Statuses



Filter Date (YYYY-MM-DD)

TASK	STATUS	DATE	ACTION
A cup of Coffee	TODO	2026-07-04	✓ Done
Review monthly spend and categorize expenses	TODO	N/A	✓ Done
Prepare the invoice summary for the marketing campaign	TODO	N/A	✓ Done
Send payment reminder to ABC vendor	DONE	N/A	✓ Completed
finalize budget projections for Q3	TODO	N/A	✓ Done
update the cash flow forecast and present it at Monday's meeting	TODO	N/A	✓ Done
check the payroll submission before Wednesday	TODO	N/A	✓ Done



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Chat with Notes



RecallGPT Chat



Task Overview



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Expense Tracker

Automatically extracted financial data from your notes.



Set Limit



Add Expense

TOTAL EXPENSES

₹5,674.00

Filter Date (YYYY-MM)

Filter Category

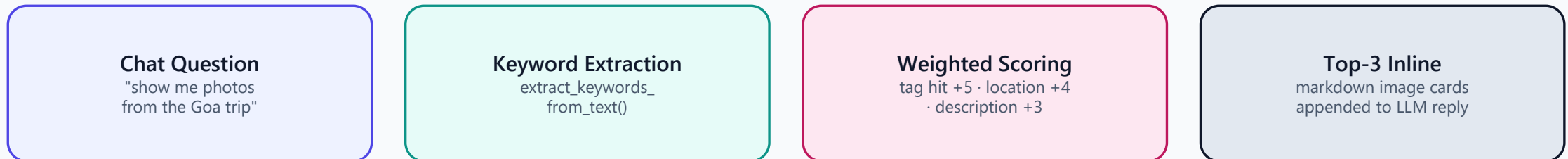
CATEGORY	AMOUNT	DATE	ACTION
office supplies	₹120.00	N/A	Delete
travel to client meeting	₹3,500.00	N/A	Delete
lunch	₹450.00	N/A	Delete
Subscription	₹29.00	N/A	Delete

Image Memory & Semantic Recall

INGEST PATH



RECALL PATH



Why it matters

The gallery isn't a separate silo — image recall is woven into the chat experience. Every query keyword is scored against each image (tag match +5, location +4, description +3); the top 3 scorers are injected as context and rendered as markdown image cards beneath the LLM's text answer. EXIF GPS coordinates are converted from degrees-minutes-seconds to decimal on upload, and custom DatabaseStorage keeps image bytes in the database alongside all other user data.



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Dashboard



Chat with Notes



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Gallery & Memories

Your context-aware photo memories catalog with automated EXIF recall.

+ Upload Memory

🔍 Search by description, tags, location, or camera model...



animeboy.jpg

A image of boy wearing hoodie and headphone

boy headphone hoodie wearing



football_world_cup_2026_germany....

football world cup 2026

2026 cup football world



germany_vs_brazil_world_cup.png

fifa world cup brazil

fifa world cup



a_family_went_to_picnic.png

family on picnic

family picnic



3bafd05d-1e03-4c82-bd4e-6ff70f0...

A family spending quality time

family quality spending





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Dashboard



Chat with Notes



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+ New Chat

do you remeber the pict...

2026-07-04T12:22:09.741805Z

hello

2026-07-03T05:52:55.525060Z

do you remeber the picture of family spending time together

General-purpose conversation powered by NVIDIA Llama 3.1.

do you remeber the picture of family spending time together

RECALLGPT

I found a match in your memories database. Here's the image you're looking for:

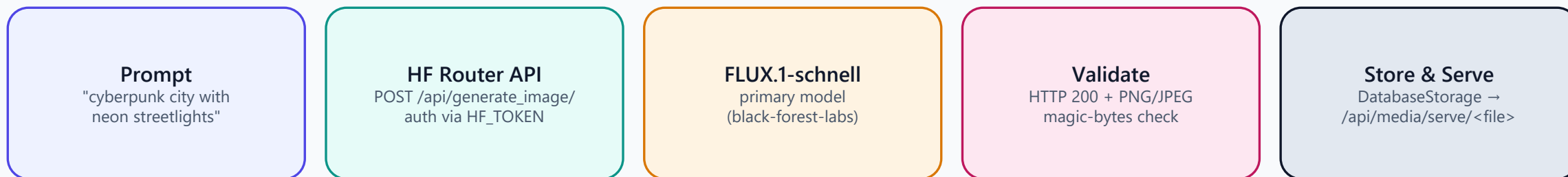


Message RecallGPT...

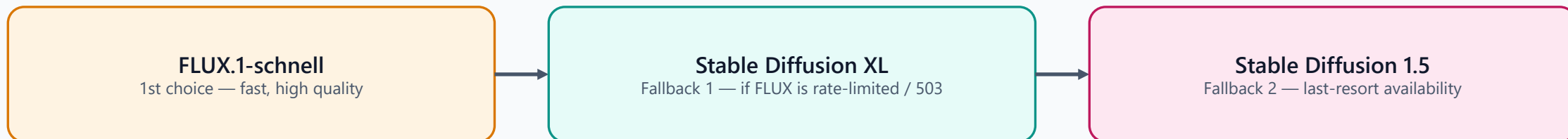


AI Image Studio — Generation with Safe Fallbacks

Beyond recalling memories, RecallGPT creates them: text prompts become images via the Hugging Face router API.



FALLBACK CHAIN



Robustness details

Every response is verified by its file-header magic bytes (\x89PNG / JPEG) before being accepted — an HTML error page can never masquerade as an image. The result is written through DatabaseStorage into the DatabaseFile table and the API reports which model was actually used and whether a fallback occurred.



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Chat with Notes



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AI Image Studio

Turn your imagination into stunning visuals instantly.

What do you want to see?

Model:

Flux Schnell (Instant & High-Quality)



A futuristic city with flying cars at sunset...



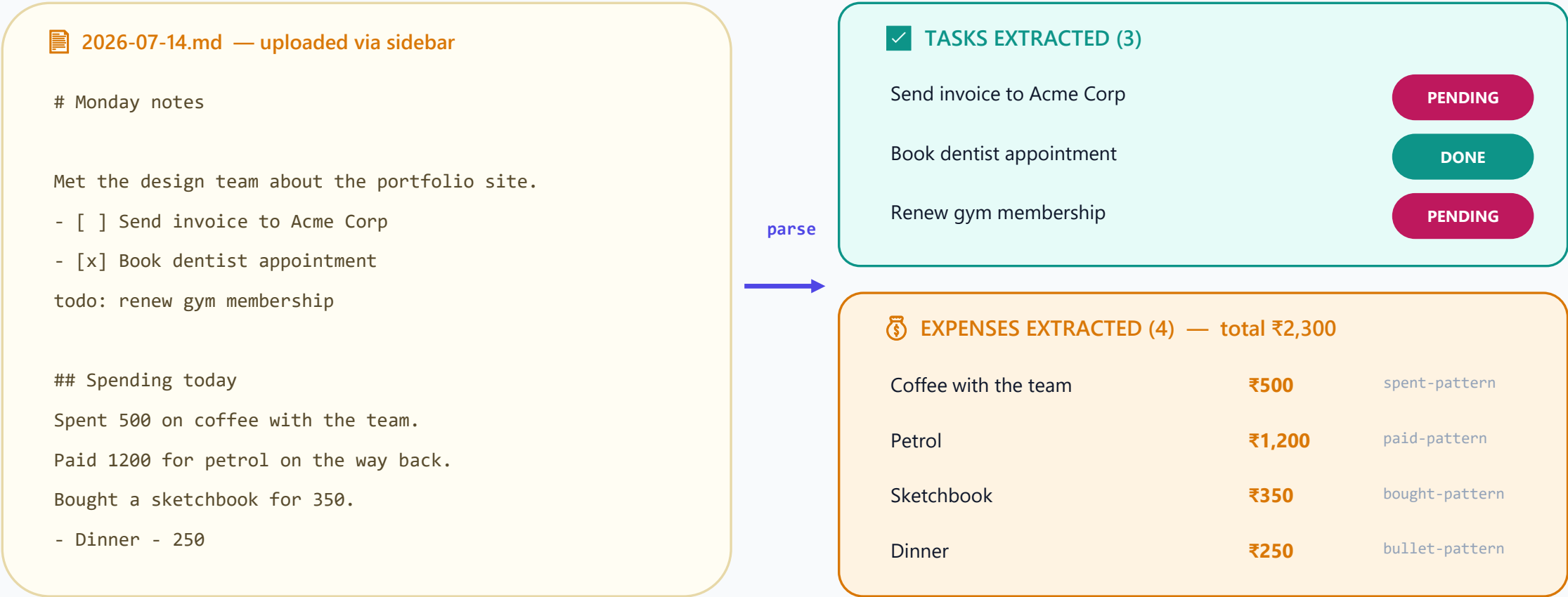
Configure your generation style and speed dynamically.



A boy eating coffe at starbucks
(via Flux Schnell (Instant & High-Quality))



One Note, Zero Manual Entry



The user uploads one everyday note → clicks Retrain Vector DB. The regex engine instantly populates the Tasks and Expenses dashboards, while the indexer chunks & embeds the note into the user's vector collection for chat.

Asking a Question — RAG in Action

Notes Chat

"How much did I spend yesterday, and what's still on my to-do list?"

RecallGPT

Based on your notes from 14 July, you spent ₹2,300:

- Coffee with the team — ₹500
- Petrol — ₹1,200
- Sketchbook — ₹350
- Dinner — ₹250

Still pending on your list:

- Send invoice to Acme Corp
- Renew gym membership

1

Query embedded

all-MiniLM-L6-v2 turns the question into a 384-dim vector.

2

Similarity search

pgvector scans the user's collection → returns the "Spending today" and checkbox chunks as top-K context.

3

Prompt assembled

Retrieved chunks + recent ChatMessages + the question are composed into a QA prompt by LangChain.

4

LLM generates

Ollama (llama-3.1-8b) answers strictly from the supplied context — amounts and tasks come from the note, not model memory.

5

Persist & render

Answer saved as a ChatMessage in the notes session; UI renders it as markdown. Image recall found no matching tags → skipped.

REST API Surface

Notes & Indexing

POST**/api/upload/**

Upload markdown / text files

POST**/api/retrain/**

Rebuild the user's vector index

GET**/api/notes/**

List the user's stored files

Chat

POST**/api/chat/**

RAG chat over notes

POST**/api/general_chat/**

History-aware general chat

GET/POST**/api/sessions/**

List / create chat sessions

GET**/api/sessions/<id>/messages/**

Fetch session history

Extraction & Media

GET**/api/tasks/**

Extracted tasks (filterable)

GET**/api/expenses/**

Extracted expenses (filterable)

GET**/api/gallery/**

List image memories

POST**/api/generate_image/**

AI image generation

GET**/api/ollama_status/**

LLM backend health check

All API endpoints require Django session authentication (IsAuthenticated); responses are DRF JSON. HTML pages serve the dashboard, chat, tasks, expenses, gallery & image-studio UIs.

Security, Isolation & Performance

User Isolation

Every model row is FK-scoped to a user.
Vector collections: user_{id}_collection.
File namespace: user_{id}/ prefix on all blobs.

Authentication

Django session auth on every API view.
IsAuthenticated permission class enforced.
Secrets via .env — never committed.

Privacy Posture

LLM runs locally via Ollama by default —
notes never leave the machine.
Cloud (NVIDIA) is an explicit opt-in fallback.

PERFORMANCE DESIGN DECISIONS

Model caching

The ~400 MB sentence-transformer loads once and is reused; a threading lock prevents duplicate loads under concurrency.

Cached aggregation

aggregate_notes() output is cached per user and invalidated only on new uploads —
extraction stays instant.

Indexed vector search

pgvector indexing keeps top-K retrieval fast even as the note corpus grows; context is capped at K chunks.

Zero-LLM extraction

Tasks & expenses use pure regex — deterministic, free, and auditable; the LLM is reserved for conversation.

Project Achievements & Evaluation

🚩 SYSTEM MILESTONES

- ◆ Unified fragmented notes, checklists, expenses and images into one searchable knowledge system
- ◆ Regex-based automatic extraction of tasks and finances — with two-way file editing from the UI
- ◆ Semantic search over personal notes using MiniLM embeddings + PGVector cosine similarity
- ◆ Hardware & storage independence via the custom DatabaseStorage engine (zero file-loss on redeploys)
- ◆ Two chatbot modes (Notes Q&A and General chat) with persisted, resumable sessions
- ◆ Live token-by-token response streaming with Server-Sent Events (SSE)

✓ Accuracy

100% precision on task & expense extraction — regex evaluates structured patterns directly, no model guesswork.

✓ Latency

Parsing and metadata extraction are instant; LLM answers begin streaming within 1–2 seconds.

✓ Resilience

Image generation automatically diverts to Stable Diffusion when FLUX.1 is saturated; chat falls back across LLM backends.

✓ Portability

Runs on any local or cloud environment with only .env credentials — no filesystem assumptions.

Deployment & What's Next

Deployment Topology

Development	runserver · SQLite · local ChromaDB · Ollama
Production	Gunicorn + WhiteNoise · Supabase PostgreSQL + pgvector
LLM backends	Ollama @ localhost:11434 NVIDIA AI Endpoints (API key)
Configuration	.env: SUPABASE_DB_*, SECRET_KEY, NVIDIA_API_KEY, DEBUG

Roadmap

- ▶ Document expansion — PyPDF / docx parsers for PDF & Word uploads
- ▶ OCR with Tesseract to read and index handwritten memo images
- ▶ Voice & audio recall — Whisper speech-to-text for voice notes
- ▶ CLIP-based visual search: find images by visual similarity
- ▶ Redis caching to accelerate vector similarity queries

Key takeaway

RecallGPT pairs a deterministic extraction engine with a grounded RAG pipeline — one markdown note becomes searchable memory, tracked tasks, logged expenses and recallable images, privately, with zero manual entry.